Second Review

ASPIRE360: AI-POWERED CAREER PATHWAYS AND INTERACTIVE SIMULATIONS

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"Choose a career that makes you excited to get out of bed every morning, where passion meets purpose, and every challenge fuels your growth."

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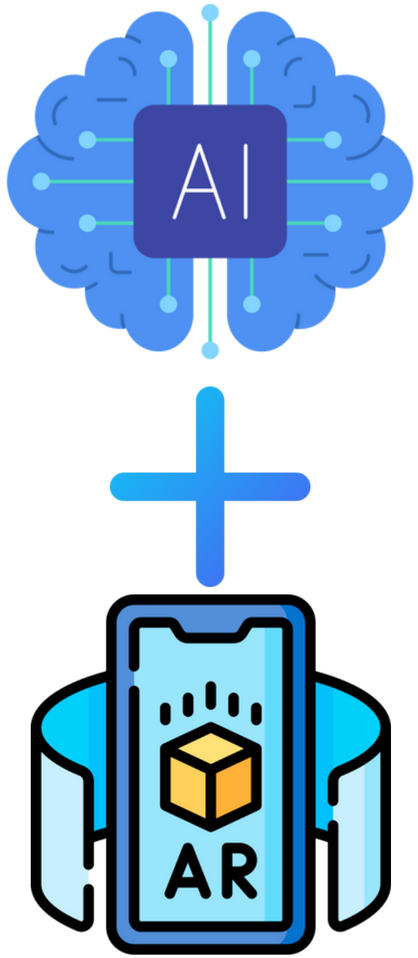
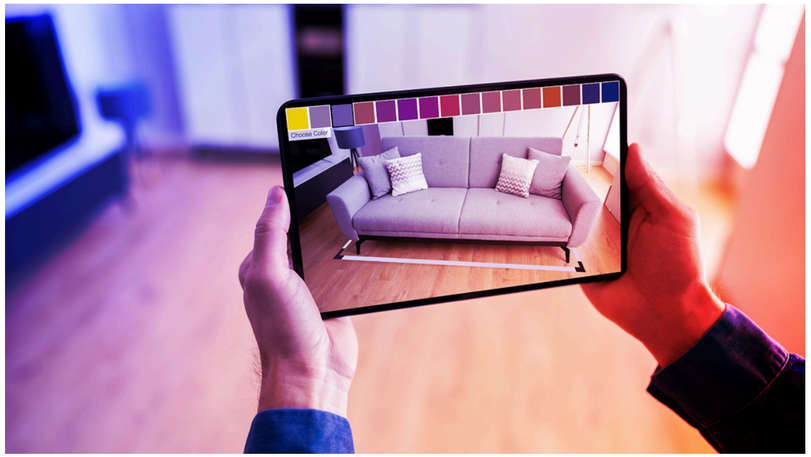
ABSTRACT

This project aims to develop an AI-driven career counseling platform that suggests suitable career options for students based on individual profiles and interests. Leveraging AI for personalized recommendations, the platform analyzes factors such as students' interests, strengths, and preferred work environments. Additionally, the platform integrates Augmented Reality (AR) to simulate real-world career environments, allowing students to experience potential career paths interactively. By combining AI and AR, this innovative approach promotes informed decision-making and holistic career exploration among students. The platform seeks to improve awareness and alignment between students' aspirations and career paths, ultimately supporting better educational and career outcomes.





INTRODUCTION

INDIA TODAY

The issue of poor career choices among students is quite widespread. Studies indicate that as much as 90% of students end up pursuing careers that aren't aligned with their skills or interests, mainly due to a lack of adequate career guidance. This misalignment can lead to dissatisfaction, underperformance, and in some cases, mental health issues like depression and stress

Moreover, a recent survey revealed that around 93% of students are only aware of a few career options such as engineering, medicine, law, and IT, overlooking the vast array of over 800 available career paths. This limited exposure, coupled with peer pressure and societal expectations, forces many students to make choices based on external influences rather than their true interests and aptitudes.

These factors contribute to high levels of job dissatisfaction and unemployment. Approximately 75% of organizations report a skill gap in the workforce, exacerbating the problem. With career counselling being insufficient, there is a pressing need for a more structured and comprehensive approach to guide students in making informed career decisions, as highlighted by experts.

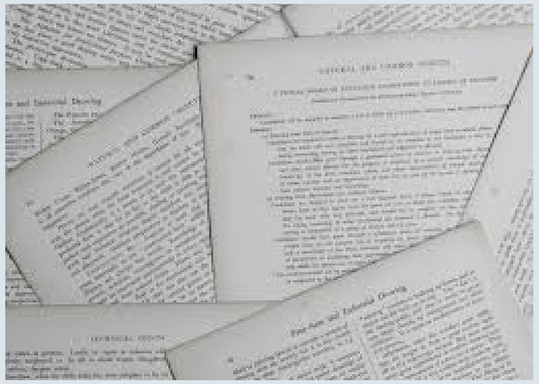
90%
Students

1.4 M
Career Counsellors

800+
Career Paths

75%
Organisations

literature SURVEY



Research Papers

Websites

Applications

Other Online Resources

RESEARCH GAP

AI-driven career guidance platforms have enhanced the process of matching users with suitable career paths by analyzing their skills, interests, and backgrounds. These systems have improved recommendation accuracy, yet they still rely largely on self-assessment data, which can be prone to biases as users may not always accurately assess their own strengths and preferences. This reliance on **subjective input can impact the precision** of recommendations, highlighting the need for more objective data integration within **AI-guided career systems.**

These innovations also offer users the chance to explore a wide range of **unconventional yet trending career options that are often overlooked.** Addressing these gaps could significantly improve the accuracy, engagement, and user experience of career guidance platforms, making them more comprehensive and impactful.

Two promising opportunities that remain underutilized:

Incorporating objective data into AI recommendations

- By integrating standardized skill assessments, AI systems could reduce reliance on self-reported data, creating more accurate and reliable career matches.

Leveraging Augmented Reality (AR) to enrich career exploration.

- AR has the potential to transform career guidance by offering immersive experiences that allow users to "test-drive" different roles, providing a deeper understanding of various career environments.

problem STATEMENT

A fulfilling career begins with informed choices

The Economic Times

Inadequate career counseling in schools leads to poor career choices among students, often resulting in job dissatisfaction, mismatched skills, and unemployment. Limited exposure to diverse career options, societal pressures, and a lack of personalized guidance contribute to this issue, especially in regions like India. To address this, there is a need for a dynamic, technology-driven solution that provides personalized career guidance, enabling students to make informed choices aligned with their skills and interests.

OUR PROCESS

1 Problem Understanding and Planning

- Business Objectives
  - Define problem and goals
  - Set project scope and objectives
  - Plan team roles and resources
  - Data Collection
- Product Development

2 Data Collection

- Market Survey
- Gather career data: skills, qualifications, growth opportunities
- Collect user data: student interests, academic performance (anonymized)

3 User Research & Prospecting

- Identify Customers
- Develop student profiles to guide AI training

4 AI Model Development

- Preprocess data and build career recommendation model
- Product Refinement
  - *Train using classification and clustering algorithms*
  - *Validate model accuracy with test data*

5 AR Development

6 Integration

- Identify core tasks for career simulations
- Develop immersive AR environments
- Add interactive elements for experiential learning
- Connect AI model with AR simulations
- Link user profiles with career simulations based on AI output

7 Testing and Feedback Collection

- *Conduct pilot testing for model recommendations*
- *Collect user feedback on AR experience and career fit*

8 Refinement and Iteration

- Improve AI model and AR components based on feedback
- Update UI/UX for enhanced user experience

9 Deployment and Monitoring

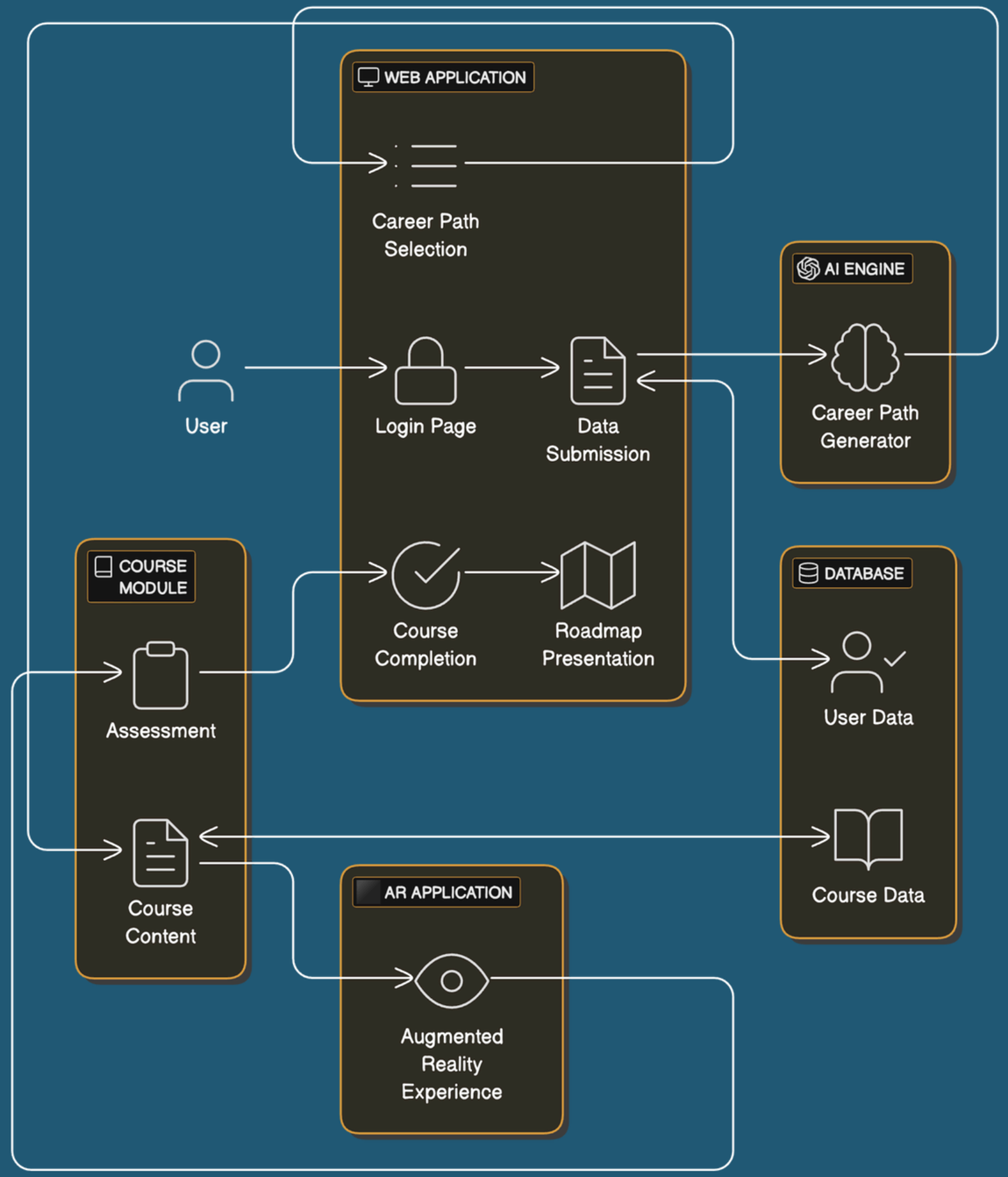
- Launch & Collect Feedback
- After Release Updates

Proposed METHODOLOGY

This methodology proposes a novel approach to enhance AI-driven career guidance systems by addressing the existing research gaps and leveraging emerging technologies. The primary focus is to integrate objective skill assessments and immersive technologies like Augmented Reality (AR) to create more accurate, reliable, and engaging career guidance experiences.

Objective : To design an AI-powered career guidance platform that integrates standardized skill assessments and AR-based immersive career exploration to improve the accuracy and user experience of career recommendations.

System Architecture



ROADMAP

"The future belongs to those who believe in the beauty of their dreams."
- Eleanor Roosevelt

Conclusion

In conclusion, the widespread issue of poor career choices among students stems from a lack of adequate career guidance, limited exposure to diverse career options, and societal pressures. These factors result in a misalignment between students' skills, interests, and chosen professions, leading to dissatisfaction, underperformance, and even mental health challenges. The skill gap reported by organizations further underscores the urgency of addressing this problem. To empower students to make informed decisions, a structured and comprehensive approach to career counselling is essential. By providing personalized guidance and raising awareness of the diverse career opportunities available, we can help students align their aspirations with their potential, fostering a more satisfied and capable workforce for the future.

project REFERENCES

- Envisioning Tomorrow: AI Powered Career Counselling
- AI-Powered Academic Guidance and Counseling System Based on Student Profile and Interests
- AI based SSS for Educations System
- Artificial Intelligence for Career Guidance – Current Requirements and Prospects for the Future
- D-Learning: An Experimental Approach to Determining Student Learning Outcomes Using Augmented Reality (AR) Technology
- Evaluating Students' Acceptance Intention of Augmented Reality in Automation Systems Using the Technology Acceptance Model
- A Study of The Use of Augmented Reality in Learning: Impacts on Increasing Students' Critical Thinking Skills

THANKS FOR THE REVIEW

"Empowering students to choose the right path today lays the foundation for a brighter, more purposeful tomorrow."